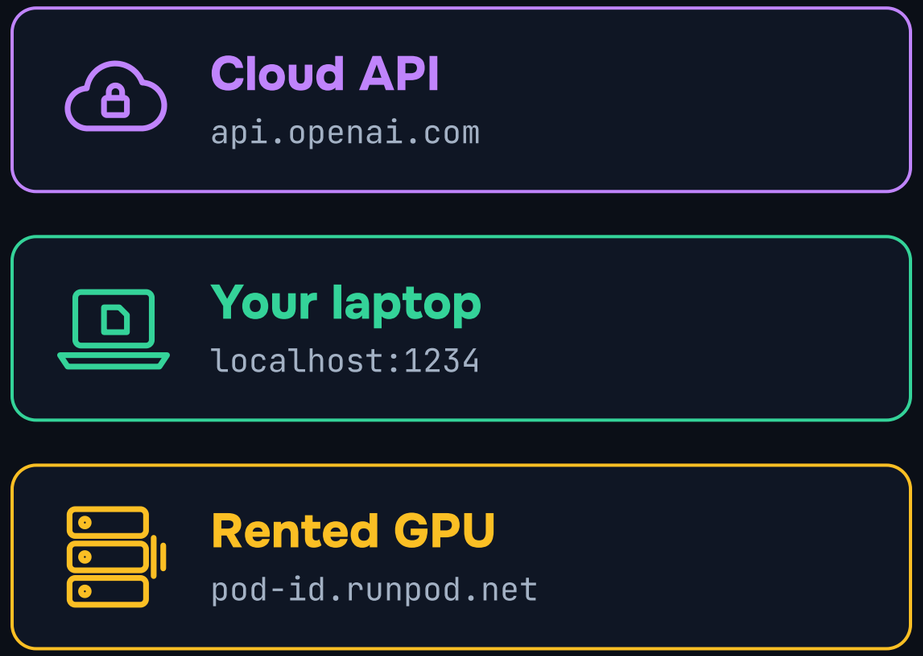
GENAI BASICS • LESSON 10 • MODULE 4: OPEN MODELS

Open models

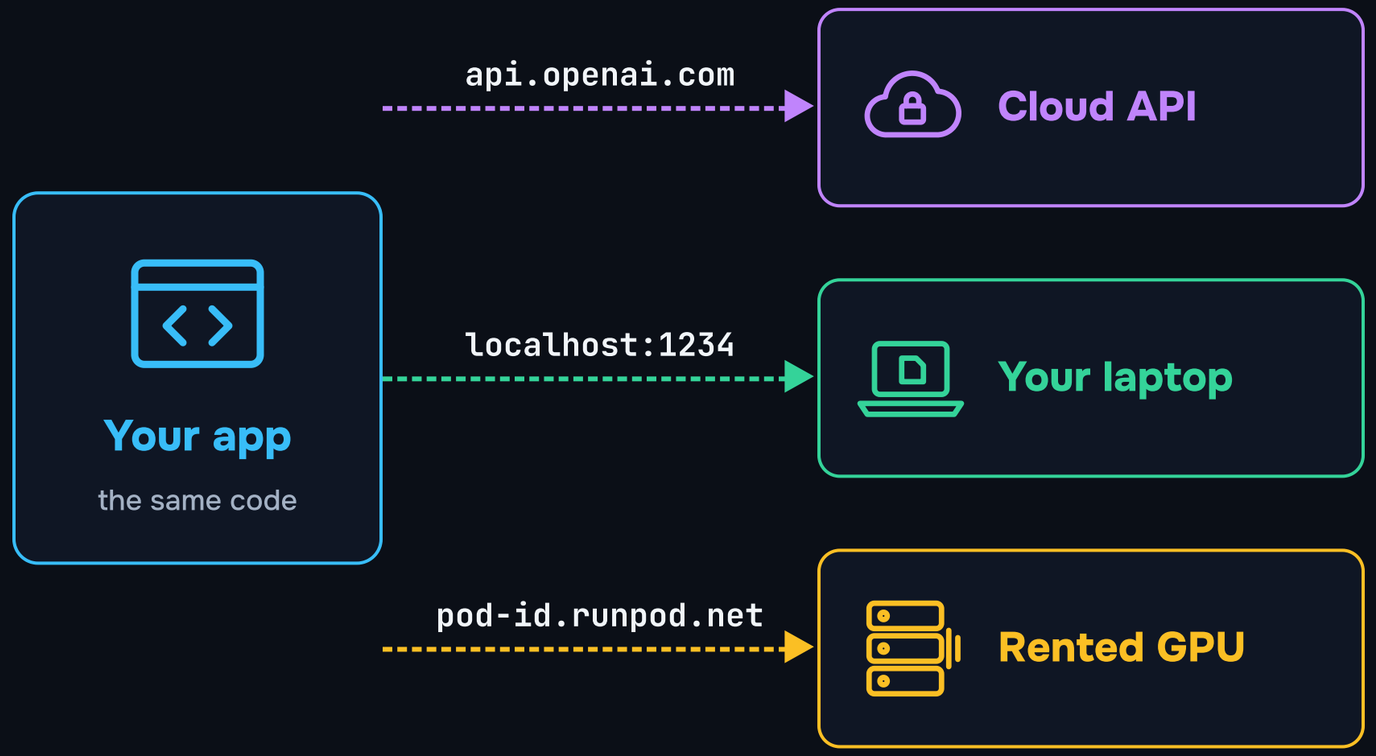
How to run a language model on your laptop and on a rented GPU

Hugging Face, LM Studio, Ollama, Runpod. Model size, memory, speed, price. Data as of October 2026.



THE IDEA

One app, three places where the model runs



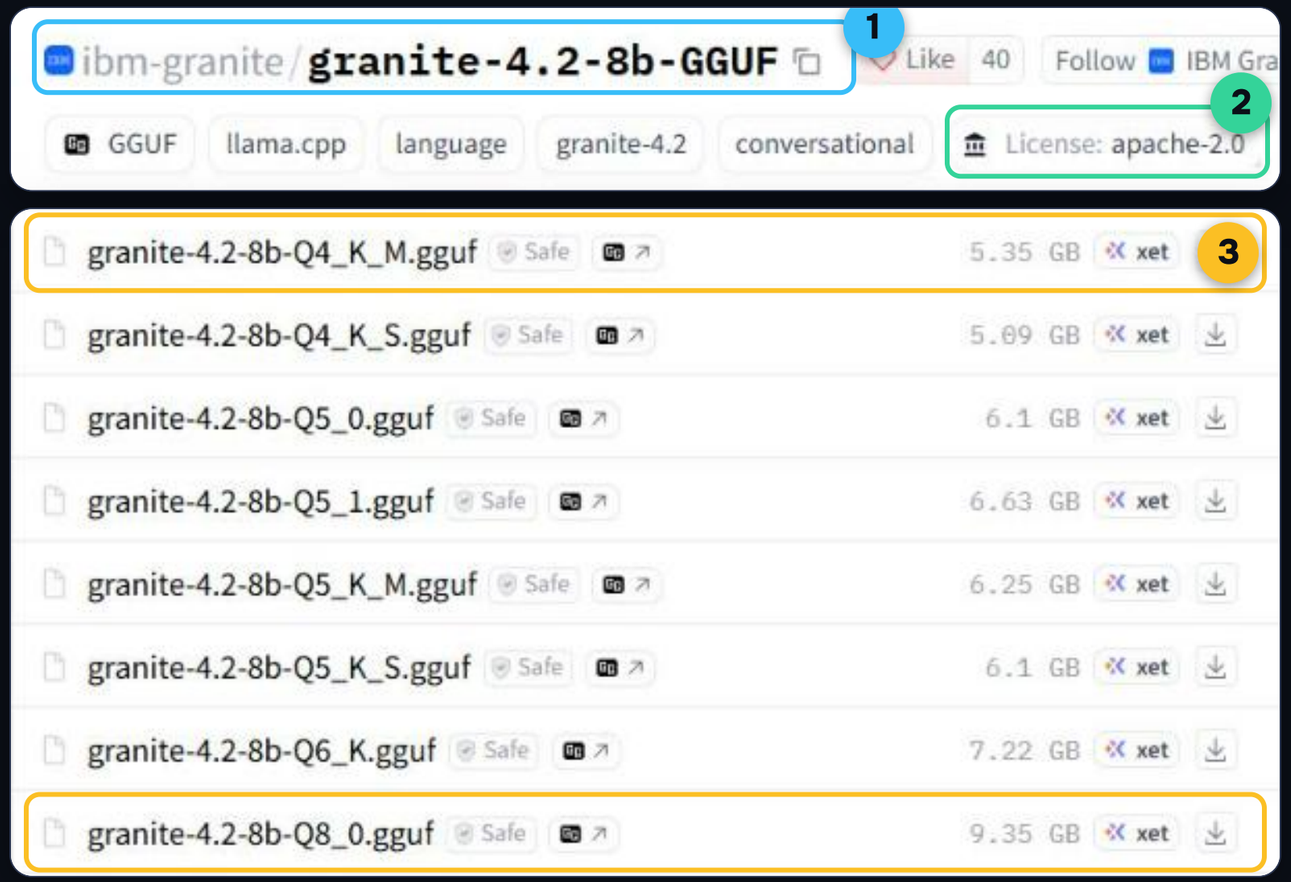
1. **Cloud API**: the vendor runs the model; you pay per token and your data goes to them
2. **Your laptop** (localhost = “this computer”): the model file is yours, no internet needed, you pay with hardware
3. **Rented GPU**: your model on someone else’s powerful machine, paid by the hour

> **Key point:** the app’s code barely changes: only the address and model name change.

OPEN MODEL

An open model is a file you can download, plus its licence

HUGGING FACE · IBM-GRANITE/GRANITE-4.2-8B-GGUF · CAPTURED 16.10.2026



1. **Who made it**: IBM, Granite 4.2 8B. **8B** = 8 billion parameters, the numbers inside the model
2. **Licence** Apache-2.0: fine for work and commercial use
3. **.gguf files** (the format for LM Studio and Ollama): one model in several sizes, here about 5–9 GB; full list on slide 6

Hugging Face is the main “warehouse” of open models. Each has a card: what it does, its limits, its licence.

“Open” ≠ “anything goes”. Llama 4 has its own restrictive licence; MusicGen (L06) is non-commercial only.

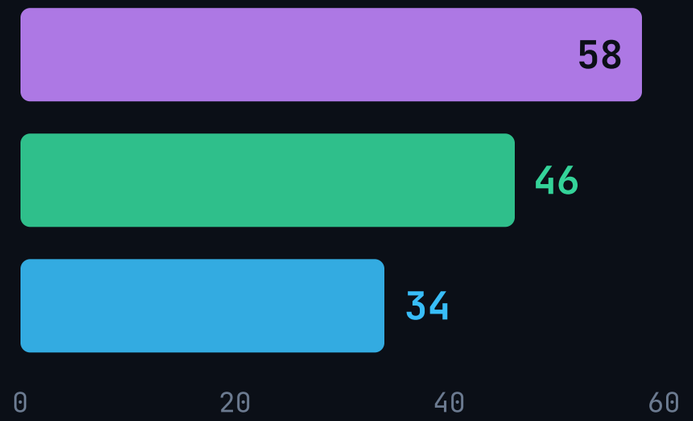
WHY

Your own model: data stays home, but quality trails the leaders

Pros: data never leaves the machine · works offline · no per-token bill · nobody changes the model behind your back

Cons: weaker than the best cloud models · needs memory and a GPU · setup and updates are your job

ARTIFICIAL ANALYSIS INTELLIGENCE INDEX, OCTOBER 2026



Higher is better. The gap between the best closed and best open model is ≈ 12 points.

A LOCAL MODEL ON WORK TASKS

✓ sorting emails and tickets

✓ drafts

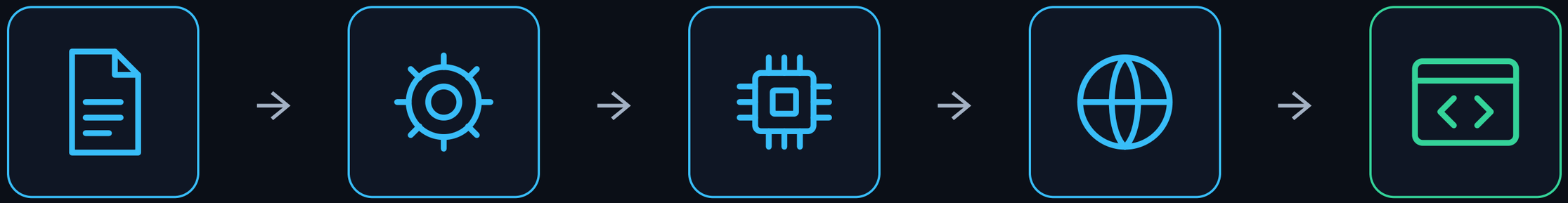
✓ search over your own documents

✗ hard reasoning, large codebases

HOW IT FITS TOGETHER

Five parts: file, engine, hardware, server, app

LM Studio or Ollama does this for you



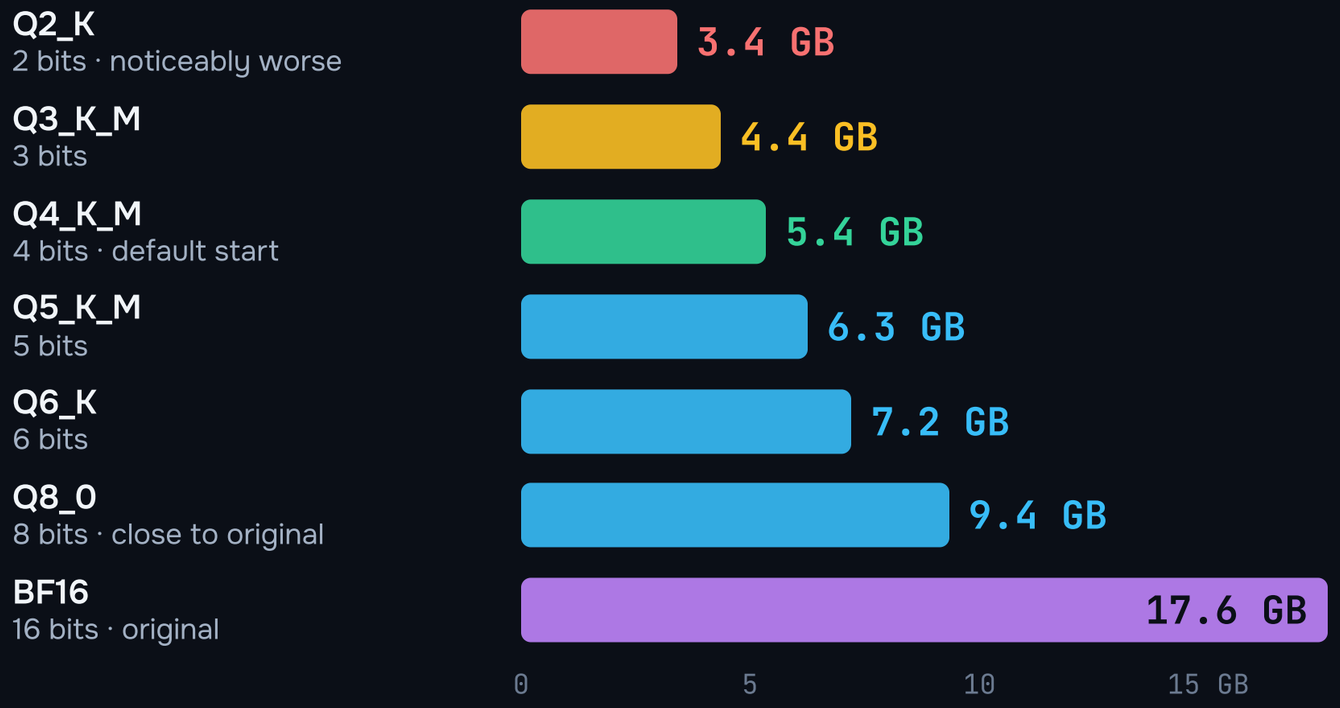
The engine (llama.cpp) reads the model file and computes the answer. **The server** gives it an address, just like a cloud API.

If it is slow, find which part is to blame: file too big? no GPU? Not “the model is bad”.

FILE SIZE

Quantisation: the same model 2–5× smaller, slightly lower quality

GRANITE 4.2 8B: FILE SIZES ON HUGGING FACE



Q4

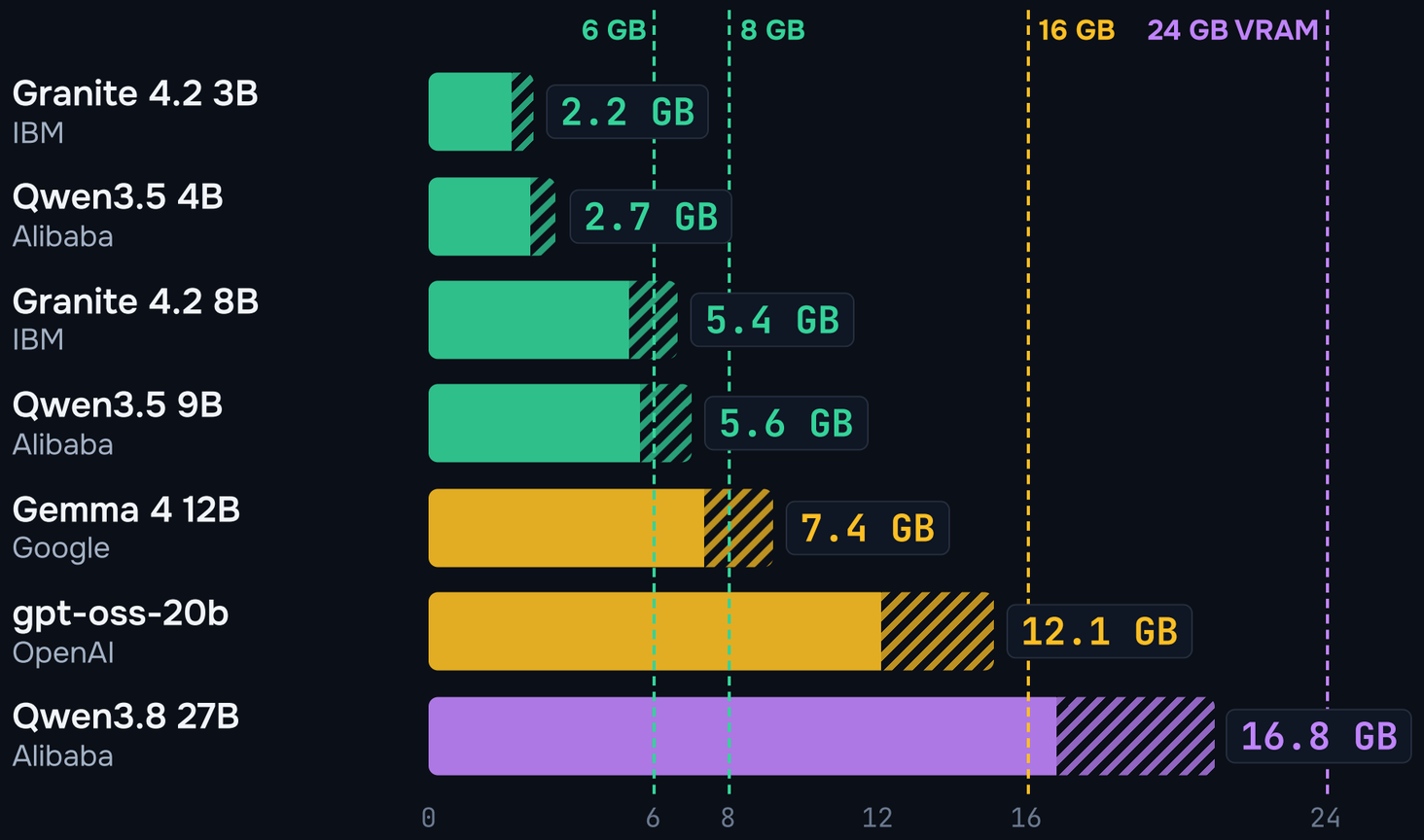
start with 4 bits; the LM Studio docs say the same

> **How it works:** each number in the model is stored more coarsely, with fewer bits. Like JPEG for photos: a smaller file, slightly fewer details.

WILL IT FIT

The model file plus room for the conversation must fit the GPU memory

Q4_K_M FILE + 25% MARGIN (HATCHED) · COLOUR = THE SMALLEST GPU IT FITS



+25%

on top of the file: memory for the context, the conversation history (usually 20–30%)

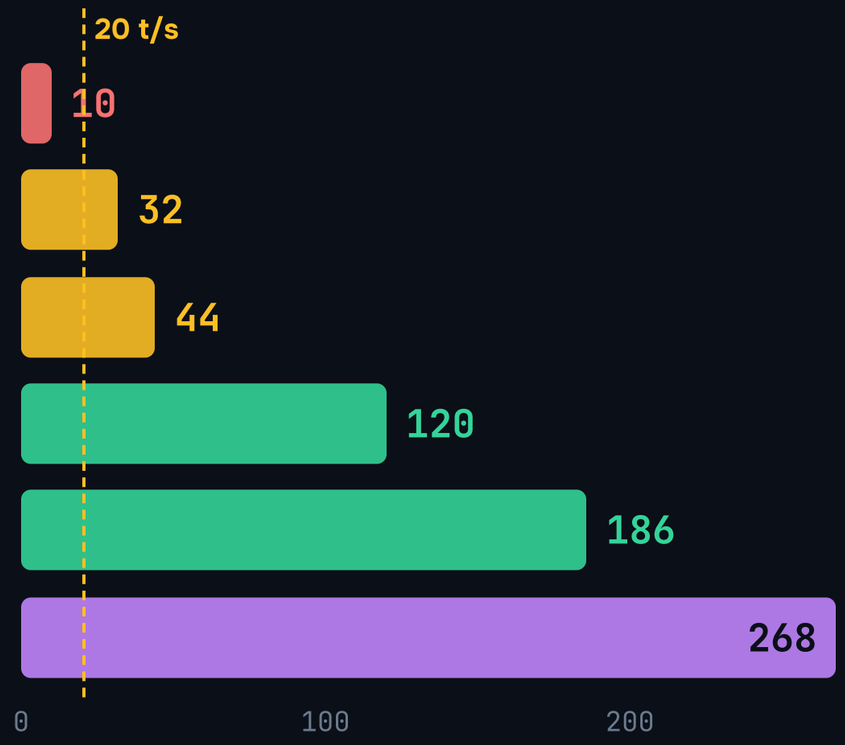
GPU memory (VRAM): **6 GB** → up to 4B · **8 GB** → up to 9B · **16 GB** → up to 12B (20B just fits) · **24 GB** → up to 27B.

Without a GPU the model lives in system RAM: it fits, but runs slowly. Take 3–4B.

SPEED

A GPU answers 2–25× faster than a CPU

Tokens per second (a token ≈ part of a word), 7B model at 4 bits, llama.cpp. People read ≈ 4–7 tokens per second.



≥ 20 t/s

comfortable: long answers without waiting; 10 t/s is bearable only for short ones

> Numbers are approximate: they depend on the model, conversation length and settings. LM Studio shows your own speed under each answer.

HOW LONG YOU WAIT FOR A HALF-PAGE ANSWER (≈ 300 TOKENS)

30 s
CPU

9 s
Apple M5

7 s
RTX 4050 Laptop

1.6 s
RTX 4090

LM Studio: a chat like ChatGPT, but the model runs on your machine

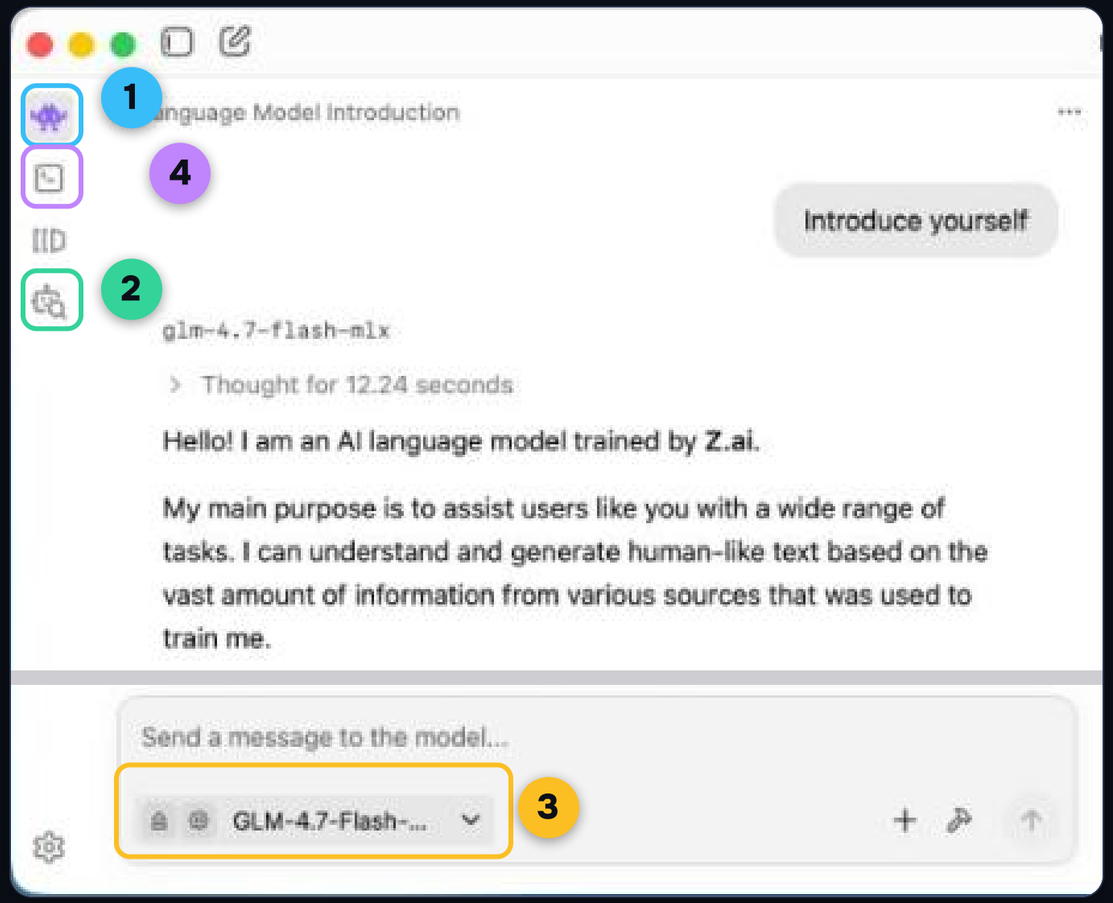


Image: LM Studio / Element Labs, 0.4.0 release blog (detail: the left chat)

1. **Chat** — talk to the model
2. **Discover** — find and download models from Hugging Face
3. **Model picker**: which one to load into memory
4. **Developer** — the local server (slide 11)

> Free, including for work. Windows, Mac (Apple Silicon), Linux. Version 0.4.26.

DEMO 1

Demo: download a small model and ask it a work question

1. **Find** — Discover → “granite-4.2-3b”
2. **Download** — Q4_K_M · 2.2 GB
3. **Load** — pick the model in Chat
4. **Ask** — the prompt below
5. **Look** — tokens/s under the answer

PROMPT

```
Ticket: “The customer cannot reset their password
after changing phones.”
Return JSON: issue, urgency (low/medium/high),
next_step.
```

WHAT A GOOD ANSWER LOOKS LIKE (JSON)

```
{
  "issue": "password reset after a phone change",
  "urgency": "medium",
  "next_step": "verify identity and reset sign-in
manually"
}
```

> **What to notice:** the answer arrives with no internet; the speed in tokens/s is under the answer; memory use rises in Task Manager.

LOCAL SERVER

Turn on the server and the model answers your code like a cloud API

PYTHON, THE SAME CODE AS IN L03

```python
from openai import OpenAI

ticket = "Customer cannot reset password"
client = OpenAI(
    base_url="http://localhost:1234/v1",
    api_key="lm-studio")                # for the cloud: your key

r = client.chat.completions.create(
    model="granite-4.2-3b",
    messages=[{"role": "user", "content": ticket}])
print(r.choices[0].message.content)
```

THE ADDRESS AND MODEL NAME CHANGE (AND THE KEY)

Cloud (OpenAI)
`https://api.openai.com/v1`
`gpt-5.4-mini`

Laptop (LM Studio)
`http://localhost:1234/v1`
`granite-4.2-3b`

Laptop (Ollama)
`http://localhost:11434/v1`
`granite4.2:3b`

> In LM Studio: **Developer → Start server**. The address `localhost` means “this computer”.

HANDS-ON WITH AN AGENT

A coding agent installs the model for you and measures the speed

PROMPT FOR A CODING AGENT: CODEX IN CHATGPT OR ANTIGRAVITY (FREE, WITH LIMITS), CLAUDE CODE (PAID)

```
Install Ollama on this computer (official installer from
ollama.com) and run the open model granite4.2:3b.
1. Check the GPU and its memory, system RAM and free disk.
Tell me which model fits.
2. Download the model and ask it: “Briefly: what is
quantisation?”
3. Measure the speed (tokens per second) and how much memory
the model uses.
4. Show the local server address and give me an example
request in Python.
5. Expose nothing to the network; the server stays on
localhost only.
```

Ollama — a similar tool that is easy to drive with commands: `ollama run granite4.2:3b`.

The agent checks the hardware, installs the program, downloads the model and reports back: speed, memory, server address.

The agent installs software on your computer: read what it proposes to run and approve each step.

SECURITY

“Local” does not automatically mean “private”: check the whole data path

- **The model computes on your machine**
  the request text goes nowhere
- **Model downloads and updates**
  go to the internet; download from official sources
- **Plug-ins and web search**
  may send data outside
- **Server exposed to the network (not localhost)**
  others can reach the model; turn on Require Authentication (token login)

Class rule: server on `localhost` only, models only from official sources (Hugging Face, ollama.com) by known publishers, client documents only with permission.

> The model licence is part of security too: do not “accidentally” ship a non-commercial model in a product.

MARKET • OCTOBER 2026

Open models by size: what runs where

GPU up to 8 GB

- **Granite 4.2 3B · 8B**
  Apache-2.0
- **Qwen3.5 4B · 9B**
  Apache-2.0
- **Gemma 4 E4B** (≈4B, for phones and laptops)
  Apache-2.0

16–24 GB GPU

- **Gemma 4 12B**
  Apache-2.0
- **gpt-oss-20b**
  Apache-2.0
- **Qwen3.8 27B**
  Apache-2.0

Server, many GPUs

- **DeepSeek V4.1**
  MIT
- **gpt-oss-120b**
  Apache-2.0
- **Llama 4**
  custom licence

> Almost everything is Apache-2.0. To start: **Granite 4.2 3B**; smarter but needs a GPU: **Qwen3.5 9B** or **Gemma 4 12B**.

BREAK

5 minutes: grab a coffee while the model downloads

05:00

Next: renting a GPU on Runpod, price, how to choose, homework.

LAPTOP LIMITS

When a laptop is not enough: four signs

Does not fit

the model is bigger than the laptop’s memory

Too slow

10,000 emails overnight is too much for a CPU

Many users

one laptop serves one person

Needs 24/7

laptops get closed and taken home

Solution: keep the same model and the same code, but move the computing to a powerful cloud GPU. Only the address and model name change.

RUNPOD

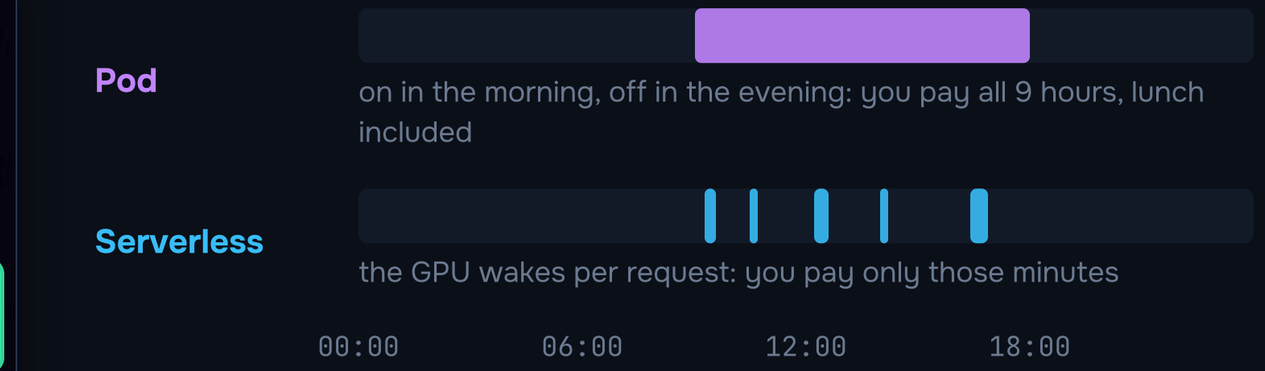
Runpod: a GPU rented with per-second billing, two modes

RUNPOD.IO/PRICING · COMMUNITY CLOUD · 24 GB · CAPTURED 16.10.2026

L4	24 GB VRAM	$0.44/hr
RTX 3090	24 GB VRAM	$0.22/hr
RTX 4090	24 GB VRAM	$0.34/hr
RTX A5000	24 GB VRAM	$0.16/hr

Green: RTX 4090, 24 GB — $0.34 per hour

ONE WORKING DAY OF A SUPPORT BOT



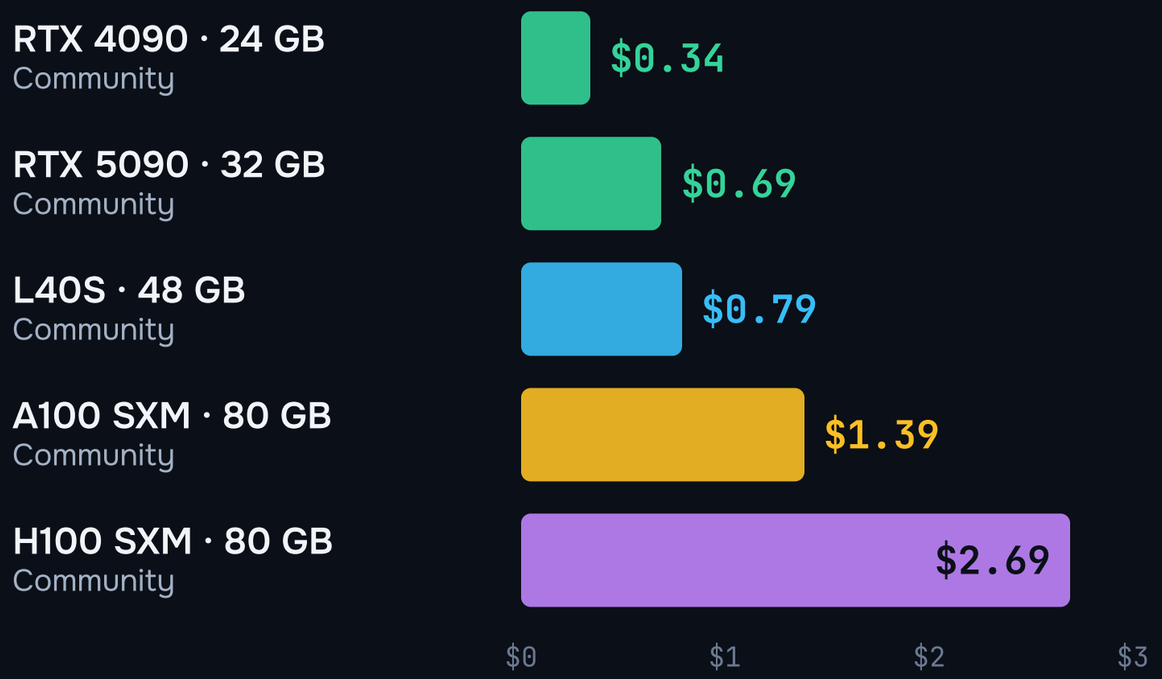
> **Trap:** a stopped Pod (Stop) still bills for its disk. When you are done, delete it (Terminate).

PRICE

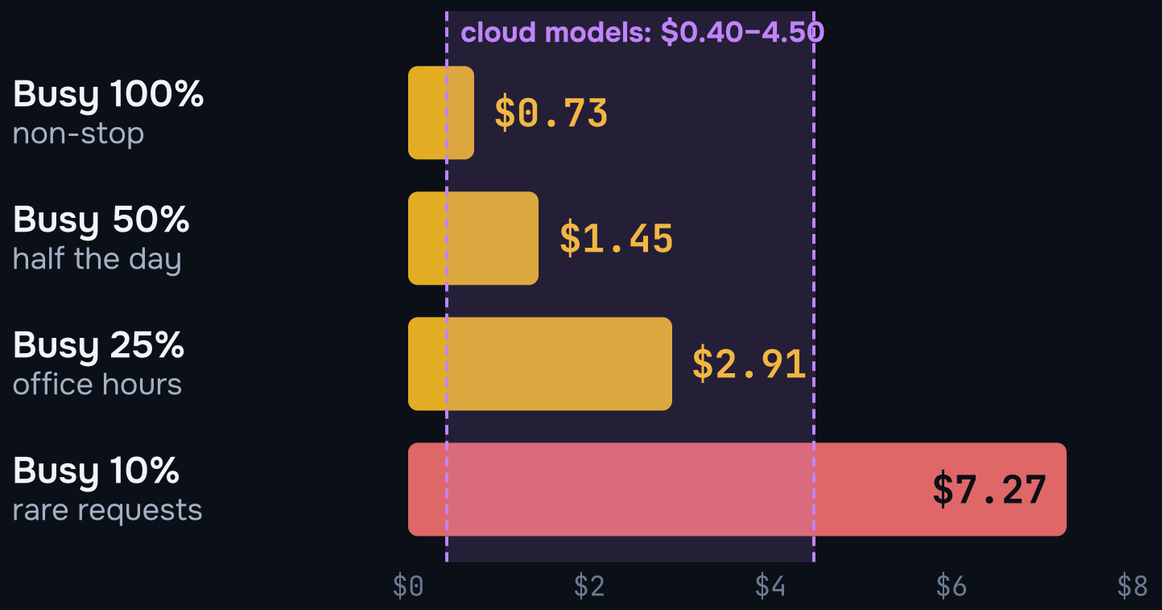
Renting a GPU only pays off if it is busy all the time

Runpod Pods, price per hour, 16.10.2026

PRICE PER HOUR



RTX 4090, 8B MODEL: PRICE OF 1M TOKENS



Band: cheap cloud models (Gemini Flash-Lite from $0.40, GPT mini up to $4.50) per 1M output tokens, nothing for idle time. Even a fully busy card costs more than the cheapest ones.

> An idle GPU costs as much as a busy one. The reason for your own model is usually privacy and control, not savings.

HOW TO CHOOSE

Choose by two questions: can the data leave, and how much work is there

	Few requests	Many requests
Data may go to the model vendor	**Cloud API** ChatGPT, Gemini, Claude: best quality, pay per request	**Cloud API** the vendor’s batch processing is usually cheaper than renting
Data must stay with you	**Your laptop** LM Studio, Ollama	**Runpod** Pod for a steady flow; Serverless for bursts

> Runpod is also someone else’s computer, but the model and data are under your control, not the model vendor’s. When you move, only the address and model name change. For sensitive data use Secure Cloud (pricier, e.g. RTX 4090 $0.89/h).

HOMEWORK

Homework: run a model yourself and compare it honestly with the cloud

1. Install LM Studio or Ollama; download a 3–9B model at Q4 (16 GB+ GPU: up to 20B)
2. Give it 3 work tasks, and the same tasks to ChatGPT or Gemini
3. Record the speed (tokens/s) and the memory the model uses
4. Score the answers 1–5 and draw a conclusion
5. Fill the card: API / laptop / Pod / Serverless and why

```
genai-homeworks/L10/
├ README.md   # choice and conclusion
├ results.md  # 3 tasks: local vs cloud
├ speed.png   # screenshot with tokens/s
└ decision.md # decision card
```

> No suitable computer? Take the smallest model (Granite 4.2 3B, 2.2 GB); it runs without a GPU.

Synthetic or your own data only. Runpod is optional and paid.

SUMMARY

Your own model is a file and memory; in code only the address and model name change

Open model = file + licence

find it on Hugging Face, read the card

Q4 + 25% must fit the GPU

otherwise it is slow; without a GPU take 3–4B

LM Studio or Ollama → localhost

only the address and model name change in code; never expose the server

Runpod: idle time costs money

when done — Terminate, not Stop

Next, L11: function calling and your first agent: the model not only answers but acts.